Name_____ Date_____ Period_____

LESSON 3-4 · HOMEWORK PACKET · SUPPORTED · SESSION 1 · PAGES 1–2

Graphing Ratios on the Coordinate Plane

Maryland Standard 6.RP.A.3a · Learning target – I can turn a ratio into ordered pairs and plot them on a coordinate plane.

REMEMBER

- Write each pair from a ratio table as an **ordered pair (x, y)**. The first quantity is x.
- Points for equivalent ratios lie on a **straight line through (0, 0)**.

WORKED EXAMPLE

A pack holds 4 juice boxes. Let x = packs and y = juice boxes. Write ordered pairs for 1, 2, and 3 packs.

Step 1 Table: 1 pack, 4 boxes; 2 packs, 8 boxes; 3 packs, 12 boxes

Step 2 Ordered pairs: (1, 4), (2, 8), (3, 12)

Answer **(0, 0), (1, 4), (2, 8), (3, 12) lie on one line.**

WORD BANK

ordered pair – (x, y): x across, then y up

origin – the point (0, 0)

x-axis – the line that goes across

y-axis – the line that goes up

PRACTICE

show your work · label units

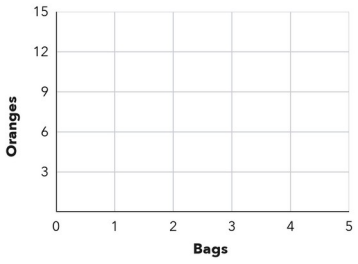
1. Each bag holds 3 oranges. Complete the table. Then write each column as an ordered pair.

Step 1 Multiply each number of bags by 3.

Step 2 Write each column as (bags, oranges).

Bags (x)	1	2	3	4
Oranges (y)	3			
(x, y)	(1, 3)			

2. Plot the ordered pairs from Exercise 1. Draw a line through the points. Does the line pass through (0, 0)? What does (0, 0) mean?



Step 1 Start at (0, 0). Go across to the x-number.

Step 2 Then go up to the y-number and draw a dot.

Step 3 Use a ruler to connect the dots.

(0, 0) means_____

3. A muffin batch uses 2 cups of flour (x = batches, y = cups). Write the ordered pair for:

a. 7 batches _____ **b.** 18 cups of flour _____

4. **Garden Rows.** Each **row** has **5 plants**. Let x = rows and y = plants.

Step 1 Multiply the rows by 5 to get y.

Step 2 Write the point as (rows, plants).

a. Write the ordered pair for 8 rows. _____

b. Write the ordered pair for 0 rows. What does it mean? _____

5. **Same Line?** Do the points (1, 5), (2, 10), and (3, 14) all lie on one straight line through (0, 0)?

☐ Yes ☐ No

6. **Find the Error.** Tickets cost $2 each. On a graph with x = tickets and y = cost, Ruby plotted (12, 6) for 6 tickets. Explain her mistake and write the correct point.

Frame *Ruby's mistake is ___. The correct point is (___, ___) because ___.*

_____

_____

CHECK YOURSELF	☐ I can do this on my own	☐ I need a little help	☐ I have a question about #___

AT HOME Make a table for something that comes in packs at home. Say two of its ordered pairs.

Name_____ Date_____ Period_____

LESSON 3-4 · HOMEWORK PACKET · SUPPORTED · SESSION 2 · PAGES 3–4

Interpreting Points on Ratio Graphs

Maryland Standard 6.RP.A.3a · Learning target – I can explain what a point on a ratio graph means in the situation it describes.

REMEMBER

- Each point (x, y) is a pair of amounts. Say **both** with units.
- The point **(1, y)** shows the **unit rate**: y ÷ x for any point. (0, 0) means zero of both.

WORKED EXAMPLE

A graph of hours worked (x) and dollars earned (y) passes through (3, 45). What does the point mean?

Step 1 (3, 45): 3 hours of work earns $45

Step 2 Unit rate: 45 ÷ 3 = 15, so (1, 15) is on the line

Answer **$15 per hour**

WORD BANK

unit rate – the amount for 1

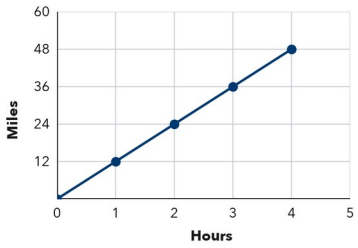
point – (x, y): x of one thing goes with y of another

on the line – y ÷ x gives the same unit rate

PRACTICE

show your work · label units

1. The graph shows time (x, hours) and distance (y, miles) for a cyclist.



Step 1 Read the x-axis label and the y-axis label.

Step 2 Find the point where x = 1. Its y is the unit rate.

a. What does the point (2, 24) mean? _____

b. What is the unit rate? _____ miles per hour

c. At this rate, how far does the cyclist ride in 6 hours? _____ miles

2. A ratio graph of trays (x) and muffins (y) passes through (1, 12). Find each missing value.

Step 1 The unit rate is 12 muffins per tray.

Step 2 To find y, multiply x by 12. To find x, divide y by 12.

a. x = 4, y = ____

b. y = 60, x = ____

c. x = 7, y = ____

d. y = 36, x = ____

e. x = 10, y = ____

f. y = 96, x = ____

3. Same Line? A ratio graph of pounds of grapes (x) and cost in dollars (y) passes through (4, 10).

Step 1 Divide 10 by 4 to find the unit rate.

Step 2 For each point, divide y by x. Does it match?

a. What is the unit rate? _____ per pound

b. Is (6, 15) on the line? Is (8, 18)? Show how you know. _____

4. Find the Error. A graph of laps (x) and minutes (y) passes through (3, 12). Omar says it means 12 laps in 3 minutes. What is wrong?

Frame *Omar is wrong because x stands for ___, so (3, 12) means ___.*

TEST PRACTICE

choose the best answer

5. A ratio graph passes through (5, 40). Which point is also on the line?

A (1, 8) **B** (6, 41) **C** (8, 5) **D** (10, 45)

6. A graph of hours (x) and dollars earned (y) passes through (2, 18). What does (1, 9) mean?

A $9 per hour **B** 9 hours for $1 **C** $18 in 1 hour **D** $2 per hour

CHECK YOURSELF	☐ I can do this on my own	☐ I need a little help	☐ I have a question about #___

AT HOME Tell someone at home what (1, 9) means on a graph of hours and dollars.